KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Integration by Parts

Name _____

S1

Integrate each expression by parts.

$$\int xe^{2x}\,dx$$

S2

Integrate each expression by parts.

$$\int x\sin(4x)\,dx$$

S3

Integrate each logarithmic expression by parts.

$$\int \ln x\,dx$$

S4

Evaluate each definite integral by parts.

$$\int_0^1 xe^{2x}\,dx$$

Show your working

KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Integration by Parts

Name _____

S1

Integrate each expression by parts.

$\int xe^{2x}\,dx$

S2

Integrate each expression by parts.

$\int x\sin(4x)\,dx$

S3

Integrate each logarithmic expression by parts.

$\int \ln x\,dx$

S4

Evaluate each definite integral by parts.

$\int_0^1 xe^{2x}\,dx$

ALG-99DB-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Parts

Name _____

E1

Integrate each expression by parts.

$$\int xe^{4x}\,dx$$

E2

Integrate each expression by parts.

$$\int x\cos(2x)\,dx$$

E3

Integrate each logarithmic expression by parts.

$$\int \ln(2x)\,dx$$

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Parts

Name _____

E1

Integrate each expression by parts.

$\int xe^{4x}\,dx$

E2

Integrate each expression by parts.

$\int x\cos(2x)\,dx$

E3

Integrate each logarithmic expression by parts.

$\int \ln(2x)\,dx$

ALG-99DB-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Parts

Name _____

E1

Integrate each expression by parts.

$\int xe^{4x}\,dx$

E2

Integrate each expression by parts.

$\int x\cos(2x)\,dx$

E3

Integrate each logarithmic expression by parts.

$\int \ln(2x)\,dx$

ALG-99DB-L3 · E

Show your working

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Integrate each expression by parts.

$\int xe^{2x}\,dx$

Answer:

$\frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + C$

Working:

Use integration by parts: $\int u\,dv = uv - \int v\,du$.

Choose $u = x$ and $dv = e^{2x}\,dx$.

Then $du = dx$ and $v = \int e^{2x}\,dx = \frac{e^{2x}}{2}$.

Substitute into the formula: $\int xe^{2x}\,dx = x \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2}\,dx$.

Take out the constant: $\int \frac{e^{2x}}{2}\,dx = \frac{1}{2}\int e^{2x}\,dx$.

Integrate: $\frac{1}{2} \cdot \frac{e^{2x}}{2}$.

Simplify: $\int \frac{e^{2x}}{2}\,dx = \frac{e^{2x}}{4}$.

Therefore $\frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S2

Integrate each expression by parts.

$$\int x \sin(4x)\, dx$$

Answer:

$$-\frac{x\cos(4x)}{4} + \frac{\sin(4x)}{16} + C$$

Working:

Use $\int u\, dv = uv - \int v\, du$.

Choose $u = x$ and $dv = \sin(4x)\, dx$.

Then $du = dx$ and $v = -\dfrac{\cos(4x)}{4}$.

Substitute: $\displaystyle\int x \sin(4x)\, dx = -\frac{x\cos(4x)}{4} - \int \left(-\frac{\cos(4x)}{4}\right) dx$.

The two minus signs give a plus.

$\dfrac{1}{4}\int \cos(4x)\, dx = \dfrac{1}{4} \cdot \dfrac{\sin(4x)}{4}$.

Simplify: $\displaystyle\frac{1}{4}\int \cos(4x)\, dx = \frac{\sin(4x)}{16}$.

Therefore $-\dfrac{x\cos(4x)}{4} + \dfrac{\sin(4x)}{16} + C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Integrate each logarithmic expression by parts.

$\int \ln x \, dx$

Answer:

$x \ln x - x + C$

Working:

Use integration by parts because the integral contains a logarithm.

Choose $u = \ln x$ and $dv = dx$.

$\frac{du}{dx} = \frac{1}{x} \cdot 1 = \frac{1}{x}$.

$du = \frac{1}{x} \, dx$.

$v = \int 1 \, dx = x$.

Substitute into $\int u \, dv = uv - \int v \, du$.

$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx$.

Simplify the remaining integral: $\int 1 \, dx = x$.

Therefore $x \ln x - x + C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Evaluate each definite integral by parts.

$$\int_0^1 xe^{2x}\,dx$$

Answer:

$$\frac{e^2(2-1)+1}{4}$$

Working:

Use $\int u\,dv = uv - \int v\,du$.

$u = x, \quad du = dx.$

$dv = e^{2x}\,dx, \quad v = \frac{e^{2x}}{2}.$

$\int xe^{2x}\,dx = x\frac{e^{2x}}{2} - \frac{1}{2}\int e^{2x}\,dx.$

$\frac{1}{2}\int e^{2x}\,dx = \frac{1}{2}\cdot\frac{e^{2x}}{2}.$

$\int xe^{2x}\,dx = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + C.$

Now use the bounds: $\left[\frac{xe^{2x}}{2} - \frac{e^{2x}}{4}\right]_0^1.$

At $x = 1$: $\frac{e^2}{2} - \frac{e^2}{4}.$

At $x = 0$: $0 - \frac{1}{4}.$

Subtract lower from upper: $\frac{e^2}{2} - \frac{e^2}{4} - (-\frac{1}{4}).$

Use a common denominator: $\frac{2e^2 - e^2 + 1}{4}.$

Factor: $\frac{e^2(2-1)+1}{4}.$

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Integrate each expression by parts.

$$\int xe^{4x}\,dx$$

Answer:

$$\frac{xe^{4x}}{4} - \frac{e^{4x}}{16} + C$$

Working:

Use integration by parts: $\int u\,dv = uv - \int v\,du$.

Choose $u = x$ and $dv = e^{4x}\,dx$.

Then $du = dx$ and $v = \int e^{4x}\,dx = \frac{e^{4x}}{4}$.

Substitute into the formula: $\int xe^{4x}\,dx = x \cdot \frac{e^{4x}}{4} - \int \frac{e^{4x}}{4}\,dx$.

Take out the constant: $\int \frac{e^{4x}}{4}\,dx = \frac{1}{4}\int e^{4x}\,dx$.

Integrate: $\frac{1}{4} \cdot \frac{e^{4x}}{4}$.

Simplify: $\int \frac{e^{4x}}{4}\,dx = \frac{e^{4x}}{16}$.

Therefore $\frac{xe^{4x}}{4} - \frac{e^{4x}}{16} + C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E2

Integrate each expression by parts.

$\int x\cos(2x)\,dx$

Answer:

$\frac{x\sin(2x)}{2}+\frac{\cos(2x)}{4}+C$

Working:

Use $\int u\,dv = uv - \int v\,du$.

Choose $u = x$ and $dv = \cos(2x)\,dx$.

Then $du = dx$ and $v = \frac{\sin(2x)}{2}$.

Substitute: $\int x\cos(2x)\,dx = \frac{x\sin(2x)}{2} - \int \frac{\sin(2x)}{2}\,dx$.

$\int \sin(2x)\,dx = -\frac{\cos(2x)}{2}$.

$-\frac{1}{2}\left(-\frac{\cos(2x)}{2}\right)$.

Simplify: $-\frac{1}{2}\int \sin(2x)\,dx = \frac{\cos(2x)}{4}$.

Therefore $\frac{x\sin(2x)}{2}+\frac{\cos(2x)}{4}+C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Parts

ALG-99DB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Integrate each logarithmic expression by parts.

$\int \ln(2x)\,dx$

Answer:

$x\ln(2x) - x + C$

Working:

Use integration by parts because the integral contains a logarithm.

Choose $u = \ln(2x)$ and $dv = dx$.

$\frac{du}{dx} = \frac{1}{2x}\cdot 2 = \frac{1}{x}$.

$du = \frac{1}{x}\,dx$.

$v = \int 1\,dx = x$.

Substitute into $\int u\,dv = uv - \int v\,du$.

$\int \ln(2x)\,dx = x\ln(2x) - \int x\cdot\frac{1}{x}\,dx$.

Simplify the remaining integral: $\int 1\,dx = x$.

Therefore $x\ln(2x) - x + C$.